

The Riemann–Hilbert Correspondence

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Contents

1	Differential Equations and Monodromy	2
2	Linear Systems and Fundamental Matrices	5

List of Figures

MEETING 1

Differential Equations and Monodromy

A differential equation is analytic, but continuation of its solutions can detect the topology of the domain. Our first example contains the essential phenomenon.

1.1 A Basic Equation on the Punctured Plane

Fix $\alpha \in \mathbb{C}$ and consider

$$y' = \frac{\alpha}{z}y, \quad z \in \mathbb{C}^\times. \quad (1.1)$$

Formally,

$$y = C \exp(\alpha \log z) = Cz^\alpha.$$

The coefficient α/z is globally defined on $\mathbb{C}^\times$, but $\log z$ is not.

If $U \subseteq \mathbb{C}^\times$ is simply connected and L is a holomorphic branch of the logarithm on U , then

$$y(z) = Ce^{\alpha L(z)}$$

is a solution. Conversely, every solution on a connected such U has this form. Thus the local solution space is one-dimensional.

Example 1.1. For $\alpha = \frac{1}{2}$, the local solutions are constant multiples of a chosen branch of $\sqrt{z}$. No branch of $\sqrt{z}$ is single-valued on all of $\mathbb{C}^\times$.

1.2 Analytic Continuation and Monodromy

Choose the base point $z_0 = 1$ and the local logarithm satisfying $L(1) = 0$. The normalized local solution is

$$y(z) = e^{\alpha L(z)}.$$

Continue it along the counterclockwise loop

$$\gamma(t) = e^{2\pi it}, \quad 0 \leq t \leq 1.$$

Along this loop, the logarithm continues from 0 to $2\pi i$. Hence, after one turn,

$$y \mapsto e^{2\pi i \alpha} y.$$

The factor $e^{2\pi i \alpha}$ is the *monodromy* around the origin.

Equivalently,

$$\exp\left(\int_\gamma \frac{\alpha}{z} dz\right) = \exp(2\pi i \alpha).$$

For a loop of winding number n , the monodromy factor is

$$e^{2\pi i n \alpha}.$$

Example 1.2. For $\alpha = \frac{1}{2}$, one turn changes $\sqrt{z}$ to $-\sqrt{z}$. For $\alpha = i$, the monodromy is $e^{-2\pi}$, so monodromy need not preserve magnitude.

1.3 The Monodromy Representation

Let $V := (\text{Sol}_X)_1$ be the space of germs of solutions at 1, given by the direct limit

$$V = \varinjlim_{U \ni 1} \text{Sol}_X(U)$$

over open neighborhoods U of 1, ordered by reverse inclusion. Analytic continuation around a loop gives an automorphism of V , and homotopic loops give the same automorphism. Thus we obtain a representation

$$\rho : \pi_1(\mathbb{C}^\times, 1) \longrightarrow \text{GL}(V).$$

Since $\pi_1(\mathbb{C}^\times, 1) \cong \mathbb{Z}$ and $\dim V = 1$, the normalized basis identifies this representation with

$$\rho(n) = e^{2\pi i n \alpha}.$$

This is the basic passage

differential equation $\longrightarrow$ local solutions $\longrightarrow$ monodromy representation.

A nonzero local solution extends to a single-valued holomorphic solution on $\mathbb{C}^\times$ precisely when

$$e^{2\pi i \alpha} = 1,$$

that is, when $\alpha \in \mathbb{Z}$. This does not imply extension across 0: for example, z^{-1} is single-valued on $\mathbb{C}^\times$ but has a pole at the puncture.

1.4 The First Inverse Problem

Given $T \in \mathbb{C}^\times$, can one construct an equation with monodromy T ? Choose $\alpha \in \mathbb{C}$ satisfying

$$e^{2\pi i \alpha} = T.$$

Then (1.1) has the required monodromy.

The choice is not unique: α and $\alpha + n$, with $n \in \mathbb{Z}$, give the same monodromy. The corresponding equations are related by the single-valued change of unknown

$$y = z^n \tilde{y}.$$

Thus monodromy does not remember a particular equation; it remembers it only up to an appropriate equivalence.

Example 1.3. To obtain monodromy -1 , take $\alpha = \frac{1}{2}$. More generally, every $\alpha \in \frac{1}{2} + \mathbb{Z}$ gives the same monodromy.

Summary

The equation $y' = \alpha y/z$ has local solutions $Ce^{\alpha L(z)}$. One counterclockwise turn around 0 changes L by $2\pi i$, so the solution is multiplied by $e^{2\pi i\alpha}$. This converts analytic continuation into a representation of the fundamental group and leads to the inverse question: which representations arise from differential equations?

Examples

1. Show that a loop of winding number n acts by $e^{2\pi i n\alpha}$.
2. For $\alpha = \frac{1}{3}$, $\sqrt{2}$, and i , compute the monodromy and determine whether it has finite order.
3. Consider

$$y' = \left(\sum_{j=1}^m \frac{\alpha_j}{z - a_j} \right) y.$$

Compute the monodromy around a small counterclockwise loop enclosing only a_j .

4. Given $T \in \mathbb{C}^\times$, describe all α such that $e^{2\pi i\alpha} = T$.

References for Meeting 1

- [CL55] Earl A. Coddington and Norman Levinson. *Theory of Ordinary Differential Equations*. New York: McGraw–Hill, 1955.
- [For81] Otto Forster. *Lectures on Riemann Surfaces*. Trans. by Bruce Gilligan. Vol. 81. Graduate Texts in Mathematics. New York: Springer-Verlag, 1981.
- [Iwa+91] Katsunori Iwasaki et al. *From Gauss to Painlevé: A Modern Theory of Special Functions*. Vol. E16. Aspects of Mathematics. Braunschweig: Friedr. Vieweg & Sohn, 1991.
- [Voi02] Claire Voisin. *Hodge Theory and Complex Algebraic Geometry I*. Trans. by Leila Schneps. Vol. 76. Cambridge Studies in Advanced Mathematics. Cambridge: Cambridge University Press, 2002.

MEETING 2

Linear Systems and Fundamental Matrices

Goals

- Prove local existence and uniqueness for $Y' = A(z)Y$; see why solutions continue along *every* path.
- Introduce fundamental matrices and Liouville’s formula.
- Define monodromy matrices and their dependence on basis and base point.
- Convert scalar equations to systems; compute a first non-diagonalizable example.

We pass from rank one to rank r :

$$Y' = A(z)Y,$$

where A is an $r \times r$ holomorphic matrix on a connected open set $X \subseteq \mathbb{C}$. The solution space is r -dimensional, so monodromy becomes a *matrix*, and no closed formula is available in general, so we argue abstractly. This is the argument that generalizes to connections on bundles.

2.1 Local Solutions

Theorem 2.1 (Existence and uniqueness). *Let A be holomorphic on a disc $D = D(z_0, R)$ and let $v \in \mathbb{C}^r$. Then there is a unique holomorphic $Y : D \rightarrow \mathbb{C}^r$ with $Y' = AY$ and $Y(z_0) = v$. On any connected open $U \ni z_0$ where A is holomorphic, a solution is determined by $Y(z_0)$.*

Proof. Existence. Fix $r < R$ and let $M = \sup_{|z-z_0| \leq r} \|A\|$. Set $Y_0 \equiv v$ and

$$Y_{n+1}(z) = v + \int_{[z_0, z]} A(\zeta) Y_n(\zeta) d\zeta.$$

Each Y_n is holomorphic (D is star-shaped), and by induction

$$\|Y_{n+1}(z) - Y_n(z)\| \leq \|v\| \frac{(M|z - z_0|)^{n+1}}{(n+1)!} \quad (|z - z_0| \leq r).$$

So Y_n converges uniformly to a holomorphic Y with $Y(z) = v + \int_{[z_0, z]} AY d\zeta$, hence $Y' = AY$. Since $r < R$ was arbitrary, Y exists on all of D .

Uniqueness. Differentiating gives $Y^{(k)} = P_k Y$ with $P_0 = I$, $P_{k+1} = P'_k + P_k A$, so the Taylor coefficients at z_0 are determined by $Y(z_0)$. Apply the identity theorem. ■

Remark 2.2 (Linearity is essential). The solution exists on the *whole* disc where A is holomorphic. Compare $y' = y^2$, whose solutions $1/(c - z)$ have poles at positions depending on the initial condition. For linear equations, singularities occur only where the coefficients are singular. This is what makes monodromy well defined.

Evaluation at z_0 identifies the space of solution germs with $\mathbb{C}^r$.

2.2 Fundamental Matrices

A *fundamental matrix* is an invertible matrix solution $\Phi' = A\Phi$; for instance, take the columns Y_j with $Y_j(z_0) = e_j$.

Lemma 2.3. *Let Φ be a fundamental matrix on a connected open set U .*

1. *Every solution is $Y = \Phi c$ for a unique $c \in \mathbb{C}^r$.*
2. *Any other fundamental matrix is $\Psi = \Phi C$ with $C \in GL_r(\mathbb{C})$ constant.*
3. *(Liouville) $(\det \Phi)' = (\operatorname{tr} A) \det \Phi$, so $\det \Phi(z) = \det \Phi(z_0) \exp \int_{z_0}^z \operatorname{tr} A$. In particular, Φ invertible at one point is invertible everywhere on U .*

Proof. We have $(\Phi^{-1})' = -\Phi^{-1}\Phi'\Phi^{-1} = -\Phi^{-1}A$, so $(\Phi^{-1}Y)' = -\Phi^{-1}AY + \Phi^{-1}AY = 0$. This gives (1), and (2) follows column by column. For (3), Jacobi's formula gives

$$(\det \Phi)' = \operatorname{tr}(\operatorname{adj}(\Phi)\Phi') = \operatorname{tr}(\operatorname{adj}(\Phi)A\Phi) = \operatorname{tr}(A\Phi \operatorname{adj}(\Phi)) = \operatorname{tr}(A) \det \Phi. \quad \blacksquare$$

A fundamental matrix is thus a choice of basis of solutions. The constant C multiplies on the right: $C\Phi$ is generally not a solution, since $(C\Phi)' = CA\Phi \neq AC\Phi$.

Remark 2.4 (Warning: $\exp \int A$). In rank one, $\exp \int A$ always works. For $r > 1$ it works when the $A(z)$ commute (e.g. A constant, $\Phi = e^{zA}$), since $(e^B)' = B'e^B$ requires $[B, B'] = 0$.

For the Airy system $A = \begin{pmatrix} 0 & 1 \\ z & 0 \end{pmatrix}$ one has $A(z)A(w) = \operatorname{diag}(w, z) \neq A(w)A(z)$, and the formula fails. This non-commutativity is the source of the difficulty of the subject.

2.3 Continuation Along Paths and Monodromy

Proposition 2.5. *Let γ be a path in X from z_0 to z_1 and Φ a fundamental matrix near z_0 . Then Φ continues uniquely along γ to a fundamental matrix Φ_γ near z_1 .*

Sketch. By compactness, cover γ by a chain of discs $D_j \subseteq X$ with consecutive discs overlapping along γ . By Theorem 2.1 solutions exist on each whole disc. Overlaps are convex, hence connected, so uniqueness patches the solutions together. Invertibility persists by Liouville. $\blacksquare$

By the monodromy theorem, Φ_γ depends only on the homotopy class of γ rel endpoints: the only obstruction to a global solution is topological. For a loop γ at z_0 , Lemma 2.3(2) gives

$$\Phi_\gamma = \Phi M_\gamma$$

for a unique $M_\gamma \in GL_r(\mathbb{C})$, the *monodromy matrix*.

Convention. Continuing ΦM_γ along δ gives $\Phi M_\delta M_\gamma$. With “ $\gamma\delta$ means γ first” we would get $M_{\gamma\delta} = M_\delta M_\gamma$, an anti-homomorphism. We therefore let $\gamma \cdot \delta$ mean “first δ , then γ ”; then

$$\rho : \pi_1(X, z_0) \rightarrow GL_r(\mathbb{C}), \quad [\gamma] \mapsto M_\gamma$$

is a homomorphism, agreeing with Meeting 1 when $r = 1$. (With the other convention use M_γ^{-1} .)

Dependence on choices. Replacing Φ by ΦC gives

$$\widetilde{M}_\gamma = C^{-1} M_\gamma C,$$

a simultaneous conjugation. Changing the base point along a path c transports bases and identifies the two representations up to isomorphism. The intrinsic object is the solution space with its $\pi_1(X)$ -action.

Continuing Liouville’s formula around γ also gives $\det M_\gamma = \exp \oint_\gamma \operatorname{tr} A$.

2.4 Scalar Equations as Systems

For $y'' + py' + qy = 0$, setting $Y = (y, y')^T$ gives

$$Y' = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} Y. \quad (2.1)$$

Since $y \mapsto (y, y')$ is an isomorphism of solution spaces compatible with continuation, both have the same monodromy. The determinant of $\Phi = \begin{pmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{pmatrix}$ is the Wronskian, and Liouville becomes Abel’s identity $W' = -pW$.

Example 2.6 (Euler equation). On $\mathbb{C}^\times$ consider $z^2 y'' + zy' - \alpha^2 y = 0$. The companion system (2.1) has a *double* pole at 0. Instead take $u = y$, $v = zy'$. Then $u' = v/z$ and $v' = y' + zy'' = \alpha^2 y/z$, so

$$Y' = \frac{R}{z} Y, \quad R = \begin{pmatrix} 0 & 1 \\ \alpha^2 & 0 \end{pmatrix},$$

with a *simple* pole. Since R is constant, $\Phi = \exp(RL(z))$ is a fundamental matrix, and one turn around 0 gives $M = \exp(2\pi i R)$.

- $\alpha \neq 0$: R has eigenvalues $\pm\alpha$, the solutions are $z^{\pm\alpha}$, and $M \sim \operatorname{diag}(e^{2\pi i\alpha}, e^{-2\pi i\alpha})$.
- $\alpha = 0$: $R = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, the solutions are 1 and $L(z)$, and $M = I + 2\pi i R = \begin{pmatrix} 1 & 2\pi i \\ 0 & 1 \end{pmatrix}$: nontrivial monodromy although both eigenvalues are 1. (This is $y'' + y'/z = 0$; we meet it again in Meeting 6.)

The monodromy eigenvalues are $e^{2\pi i\lambda}$ for λ an eigenvalue of R , so λ is determined only modulo $\mathbb{Z}$: Meeting 1 in higher rank, returning in Meeting 9.

The double pole of the companion system was only apparent: the gauge transformation $\operatorname{diag}(1, z)$ removes it. Pole order alone does not detect regularity; compare $y' = -y/z^2$, whose solution $e^{1/z}$ is genuinely irregular (Meeting 7).

Summary

A rank- r system has an r -dimensional solution space, and every solution continues along every path. A fundamental matrix chooses a basis; going around γ multiplies it on the right by M_γ ; changing basis or base point conjugates. The intrinsic object is the representation of $\pi_1(X)$ on the solution space. For $Y' = RY/z$ with R constant, $\Phi = z^R$ and the monodromy is $e^{2\pi i R}$.

Examples

1. Find the fundamental matrix with $\Phi(0) = I$ for $Y' = \text{diag}(1, -1)Y$ and for $Y' = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} Y$.
2. Compute the monodromy of $Y' = \frac{1}{z} \text{diag}(\frac{1}{2}, \frac{1}{3})Y$ around a loop of winding number n .
3. Verify Example 2.6 and compute M for $\alpha = \frac{1}{2}$ and $\alpha = 1$. When is $M = I$? Are $z^{\pm\alpha}$ holomorphic at 0 then?
4. Show that $(\Phi^T)^{-1}$ is a fundamental matrix for the dual system $Y' = -A^T Y$, with monodromy $(M_\gamma^{-1})^T$.
5. Show $\det M_\gamma = \exp \oint_\gamma \text{tr } A$, and prove that the monodromy representations at two base points are isomorphic.
6. Give a holomorphic 2×2 system on $\mathbb{C}$ for which $\exp \int_0^z A$ is not a fundamental matrix.

References for Meeting 2

- [CL55] Earl A. Coddington and Norman Levinson. *Theory of Ordinary Differential Equations*. New York: McGraw–Hill, 1955.
- [For81] Otto Forster. *Lectures on Riemann Surfaces*. Trans. by Bruce Gilligan. Vol. 81. Graduate Texts in Mathematics. New York: Springer-Verlag, 1981.
- [Iwa+91] Katsunori Iwasaki et al. *From Gauss to Painlevé: A Modern Theory of Special Functions*. Vol. E16. Aspects of Mathematics. Braunschweig: Friedr. Vieweg & Sohn, 1991.
- [Voi02] Claire Voisin. *Hodge Theory and Complex Algebraic Geometry I*. Trans. by Leila Schneps. Vol. 76. Cambridge Studies in Advanced Mathematics. Cambridge: Cambridge University Press, 2002.